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pCrispomyces 2.0 KO in Streptomyces:

Adapted from **High-Efficiency Multiplex Genome Editing of Streptomyces Species Using an Engineered CRISPR/Cas System**. Cobb RE, Wang Y, Zhao H. *ACS Synth Biol. 2014 Dec 8.* 10.1021/sb500351f

Go to the CRISP-y web : https://crispy.secondarymetabolites.org

You can find more info about it at ''CRISPy-web: An online resource to design sgRNAs for CRISPR applications in microbes'' Kai Blin, Lasse Ebdrup Pedersen, Tilmann Weber, Sang Yup Lee.Synthetic and Systems Biotechnology 2016: doi:10.1016/j.synbio.2016.01.003

Upload a Fasta file with the regions you want to target. To avoid off target cutting upload the whole assembly.

Select Guides with a good score, and if possible cutting near sites in which GC content is not too high, so construction of repair cassette is easier to make (PCR issues).

The pCrispomyces vector has the gRNA scaffold already cloned in. We would only need to include the spacer sequence. For that:

Design two 24 nt oligonucleotides (4 nt 5' sticky end + 20 nt spacer sequence) with the sticky ends being: ACGC on the forward primer and AAAC on the reverse primer.

For example, if the spacer sequence is CTCACGGACGGAGACCAGGA, then the two primers are:

a. Spacer-for: 5'-ACGCCTCACGGACGGAGACCAGGA-3'

b. Spacer-rev: 5'-AAACTCCTGGTCTCCGTCCGTGAG-3'

Such that the annealed product will be:

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5'-ACGCCTCACGGACGGAGACCAGGA -3'
    3'- GAGTGCCTGCCTCTGGTCCTCAAA-5'
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Clone of our annealed oligo will be done with BbsI (sticky ends for the BbsI sites (5' to 3') are ACGC and GTTT).

Anneal spacer oligos as follows:

a. Resuspend both oligos to 100µM in water

b. Mix 5uL FOR + 5uL REV + 90uL 30mM HEPES, pH 7.8

c. Heat to 95°C for 5min, then ramp to 4°C at 0.1°C/sec

5) Insert annealed spacer to plasmid by Golden Gate assembly:

Golden Gate reaction mixture:

Backbone (pCripspomyces plasmid)	X µL 100 ng	
Insert (annealed oligos)	0.3 µL	3:1 mol ratio (dilute annealed oligos 10-fold)
T4 Ligase Buffer (NEB)	2 µL	
T4 ligase (NEB)	1 µL	400 U/µL stock is sufficient; **add last**
BbsI (NEB)	1 µL	

H_2O to 20 µL

Golden Gate Program:

1. 37°C 10 min
2. 16°C 10 min
3. Goto step 1, 9 times
4. 50°C, 5 min
5. 65°C, 20 min
6. 4°C, hold

Transform 3 µL of each reaction to 50 µL E. coli dh10b by heat shock

Plate 10% of recovery culture on selective plates with 10uL of 0.5 M IPTG and 40uL of 20mg/mL X-gal (in DMSO).

Pick white colonies to selective LB+ and recover plasmid. This should be the plasmid with the guide inserted. To add an homology template repair:

PCR amplify two homology arms from gDNA of the strain to KO. (**not including the cutting site**! Also **not including an XbaI site**) Editing template must be cloned on a XbaI site in this construct!

Design overlaps (20-30 nt) at the junction of the two arms and XbaI cutsites at the opposite ends of the two arms. Then splice the two arms by overlap-extension PCR, and digest and ligate the product into the digested plasmid.

Clone the editing template into the gRNA containing plasmid (if having a lot of recircularization, consider dephosporilating after XbaI cutting)

Verify plasmid, and deliver to Streptomyces via conjugation/electroporation.

After editing, plasmid removal can be done by incubation in non-selective medium at 37 C.

Intergeneric conjugation of DNA from *E.coli* to *Streptomyces*

Adapted from (Kieser et al., 2000)

To avoid the methyl specific restriction system in Streptomyces, the non-methylating *E.coli* strain ET12567/pUZ8008 was transformed with the required plasmid to be introduced in to the desired *Streptomyces* strain.

An overnight culture of the ET12567/pUZ8008 transformant was grown as described above and used as inoculum for a 50 mL culture in a 1-3:100 ratio and incubated at 37°C at 220 rpm in a shaking incubator until the OD600 reached 0.4-0.6. Cells were harvested by centrifugation at 4200 x g for 10 min, washed twice with LB medium to remove antibiotics and resuspended in a final volume of 500 µL LB medium. In parallel a suspension of Streptomyces spores was prepared to a concentration of 2 x 108 spores/mL in 2YT medium and heat shocked at 50°C for 10 min. The spore suspension was mixed with the *E.coli* cells (500 µL + 500 µL) and plated on MS medium. Plates were incubated at 30°C for 16-18 h, prior to plates being overlaid with a 1 mL of a solution containing 500 µg of nalixidic acid (to remove *E.coli* cells) and 1 mg of the appropriate selection marker for the incoming plasmid/cosmid. The overlay was left to dry in a laminar flow hood and further incubated until primary exconjugant Streptomyces colonies were observed.